



Pratham Khatkar <pratham.khatkar2003@gmail.com>

2026 IEEE 5th World Conference on Applied Intelligence and Computing : Submission (2744) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: pratham.khatkar2003@gmail.com

Tue, Apr 28, 2026 at 11:20 PM

Hello,

The following submission has been created.

Track Name: AIC2026

Paper ID: 2744

Paper Title: Adaptive OT Cybersecurity Solutions: An Integrated Zero-Trust, Digital-Twin, and Human-Centric Framework for Resilient Industrial Control Systems

Abstract:

The protection of industrial control system (ICS) has been a challenging issue within modern-day cybersecurity since these facilities lack intrinsic resistance to network-based attacks. In the light of industry 4.0 integration, the current perimeter security architecture is insufficient when dealing with attacks based on exploiting the physical effects of the tampered process command execution. In the present paper, the Adaptive OT Cybersecurity Solutions Framework (AOCSF) will be presented. This approach was developed and implemented in practice during my internship at Rockwell Automation. It consists of the following three elements: Zero-Trust access control layer based on Attribute-Based Access Control (ABAC); Digital Twin module employing physics-aware attack defense testing; Human-Centric component monitoring user behavior and conducting scenario-based incident drills. Controlled experiments were conducted with a Python-based simulation of the Tennessee Eastman Process testbed (n=50 attack attempts per category, n=15 operators). It revealed a 96% success rate of detecting False Data Injection attacks (95% confidence interval [CI]: 90.6-100%), 82% for Man-in-the-Middle (95% CI: 71.4-93%), a 87.3% decrease in authentication delay from 36.5 to 4.6 millisecond (Standard effect size $d=7.01$, $p<0.001$), and a 52.1% increases in the speed of identifying simulated incidents (from 314 seconds to 150 seconds, $d=4.59$, $p<0.001$). Removal of each individual pillar in turn and re-testing yielded significantly lower results, demonstrating that integration in key. Validation of AOCSF on the Swat water treatment testbed provided comparable detection rate, indicating the approach's scalability to different domains of application. Finally, AOCSF is compared to IEC 62443 Security levels 2 and 3 and NIST SP 800-82 Rev. 3.

Created on: Tue, 28 Apr 2026 17:50:20 GMT

Last Modified: Tue, 28 Apr 2026 17:50:20 GMT

Authors:

- pratham.khatkar2003@gmail.com (Primary)
- gurpreet.1309@chitkara.edu.in

Primary Subject Area: Intelligent Systems (Hardware & Software)

Secondary Subject Areas:
Cybersecurity

Submission Files:

2210990674_Pratham Khatkar_Research.docx (762 Kb, Tue, 28 Apr 2026 17:40:14 GMT)

Submission Questions Response:

1. Conflict of Interest
Agreement accepted
2. Status of using third-party material in your article.
I am not using third-party material for which formal permission is required
3. Certificate of originality
Agreement accepted

4. Corresponding Author's Contact number
+91 9319559501
5. Country Name
India
6. Authors Contributions
Pratham Khatkar: Study Conception & Design, Data Collection, Interpretation of Results;
Gurpreet Singh-Supervision & Guidance
7. Conference Rules and Guidelines
Agreement accepted

Thanks,
CMT team.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One [Microsoft Way](#)
[Redmond, WA 98052](#)